

Milin Patel

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EDUCATION

VIT Bhopal University

B.Tech in Computer Science (AI & ML) | CGPA: 8.60

Bhopal, MP

Sep. 2023 – Present

TECHNICAL SKILLS

Languages: Python, Java, C++, SQL

AI/ML: PyTorch, Transformers, LangChain, Hugging Face, IBM watsonx.ai, RAG, NLP, scikit-learn

Frameworks: Flask, Streamlit

Tools: Git, GitHub, Jupyter Notebook, Pandas, NumPy, REST APIs, Feature Engineering

PROJECTS

CHAOS AI | *Python, Flask, Google Gemini API, SpeechRecognition, pyttsx3*

Oct. 2024 – Jan. 2025

- Engineered a multimodal AI assistant by integrating 5 core modules including Google Gemini API, SpeechRecognition, pyttsx3, browser automation, and system automation for natural voice interactions.
- Integrated 3 intelligent capabilities conversation memory, application launching, and browser automation to execute context-aware voice commands through a unified interface.
- Architected a modular Python application comprising 4 independent service layers for speech processing, LLM orchestration, automation, and response generation, improving future extensibility.

Alzheimer's Disease Detection | *PyTorch, Transformers, ResNet34, Medical Imaging, BBO*

Aug. 2025 – Apr. 2026

- Designed a Deep Temporal Transformer architecture combining ResNet34 and Transformer encoders for MRI-based Alzheimer's disease classification across four diagnostic categories.
- Optimized model hyperparameters using Brown Bear Optimization (BBO), producing an F1-score of 0.82 during patient-level evaluation.
- Established an end-to-end PyTorch training pipeline incorporating preprocessing, feature extraction, and cross-validation before submitting the research manuscript for faculty review.

QABot with RAG and LangChain | *LangChain, IBM watsonx.ai, Hugging Face Embeddings*

Sep. 2025 – Oct. 2025

- Built a Retrieval-Augmented Generation (RAG) application integrating 4 AI technologies to deliver document-based question answering.
- Engineered semantic vector retrieval, providing context-aware information from custom sources during LLM inference, reducing model hallucinations.

EXPERIENCE

Celebal Technologies

Jaipur, RJ (Remote)

Data Science Intern

Aug. 2025 – Oct. 2025

- Completed 50 hours of industry-led training covering Python programming, exploratory data analysis, supervised learning, and machine learning fundamentals.
- Executed data preprocessing and exploratory data analysis on 10 structured datasets using Pandas and NumPy, applying feature engineering principles.
- Implemented and evaluated over 25 machine learning experiments using scikit-learn to understand model performance and evaluation metrics.

LEADERSHIP & OPEN SOURCE

GirlScript Summer of Code (GSSoC) | *Open Source Contributor*

Jul. 2025 – Dec. 2025

- Completed a 3-month open-source contribution program by participating in collaborative software development using Git, GitHub, and pull request workflows.

VITB AI Innovator Hub | *Backend Contributor*

Jul. 2025 – Sep. 2025

- Contributed to backend development, implementing core application logic and supporting AI-based resume customization.

Technomech Club, VIT Bhopal | *Core Technical Member*

Sep. 2024 – Apr. 2025

- Coordinated a major UI/UX Hackathon with 100+ participants, supervising tech infrastructure, speaker onboarding, and platform verification workflows.

CERTIFICATIONS

IBM Generative AI Professional Certificate | IBM

Applied Machine Learning in Python | Michigan University

Cloud Computing | NPTEL National Certification